SkillCT: A Multidimensional Benchmark and Execution-Free Baseline for Agent Skill Assessment

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Abstract

Agent skills are modular packages of instructions, scripts, and resources that guide agents in specialized tasks and are increasingly shared through public Web repositories. Existing skill benchmarks primarily assess quality through downstream execution, measuring whether a skill improves agent performance on concrete tasks. While effective for evaluating practical utility, such execution-centered benchmarks provide limited coverage of properties such as specification quality, behavioral stability, semantic alignment, and responsibility scope. Their reliance on task-specific tools, environments, and repeated executions also constrains scalability and pre-deployment assessment. To address these limitations, we introduce **SkillCT**, a benchmark that broadens skill assessment from downstream utility to a multidimensional characterization of skill quality. SkillCT evaluates each skill across nine complementary dimensions spanning three aspects: Static Assessment captures properties of the skill specification itself; Dynamic Assessment measures whether the induced behavior remains stable across backbone models and request variations; and Human-Assisted Assessment examines behavior-chain consistency and task granularity that are difficult to infer from task outcomes alone. These dimensions provide a broader view of skill quality than execution success alone. Applying SkillCT to a public skill corpus produces a reference benchmark that supports systematic analysis of quality weaknesses and relationships across dimensions. Building on this benchmark, we propose an execution-free baseline for estimating the quality profiles of newly introduced skills. Unlike existing execution-centered approaches, this baseline estimates a skill's multidimensional quality profile without executing the skill, using retrieved reference skills from the assessed corpus. This reduces token consumption and evaluation time, enabling efficient assessment at scale. Experiments show that the baseline reliably approximates the profiles produced by the full assessment, enabling scalable skill development, selection, and repository maintenance.

Keywords

Agent Skills, Skill Assessment, Execution-Free Evaluation

1 Introduction

Agent skills are modular packages of task-specific instructions, scripts, and resources that guide agents in specialized tasks [1, 27]. This modular design allows general purpose agents to acquire specialized capabilities at inference time without modifying the underlying model [11]. As skills become an increasingly common component of agent systems, their quality has direct implications for the reliability, efficiency, and predictability of agent behavior [21], making skill quality an increasingly important concern. As public skill repositories continue to expand, systematic skill assessment becomes increasingly important for skill development, selection, and repository maintenance.

Existing benchmarks primarily assess Skill quality through downstream task execution. SkillsBench measures whether Skills improve task completion, while SWE-Skills-Bench evaluates their contribution through paired executions with and without Skills [8, 10]. SkillAudit generates capability-aligned tasks for individual Skills and collects execution evidence covering utility, efficiency, cost, and safety [24]. These approaches establish whether Skills provide practical benefits under concrete execution settings. However, task outcomes alone do not fully characterize specification quality, behavioral stability, semantic alignment, or the appropriateness of task scope. Moreover, evaluating each new Skill requires task construction, environment setup, and repeated execution, complicating consistent quality assessment across public Web repositories.

Complementary studies examine more specific properties of Skills and their induced behavior. SkillAxe uses diagnostic signals, including trigger precision, instruction compliance, and solution-path coverage, to guide Skill refinement [6]. SkillsMetric examines static signals such as dataflow patterns, import anomalies, and capability mismatches [4]. Security-oriented studies investigate vulnerabilities and malicious behaviors [13], while SkillScope and SkillGuard examine permissions and responsibility boundaries [15, 19]. SkillEval further assesses document-level properties without downstream execution [7]. These studies establish valuable assessment perspectives, motivating a framework that brings complementary evidence together into a unified quality profile.

Two related challenges therefore remain. First, how can specification properties, behavioral evidence, and judgments of semantic alignment and task scope be organized into a multidimensional assessment? Second, how can the resulting quality profiles be estimated for newly introduced Skills without repeating the full assessment process? Addressing both challenges requires a distinction between constructing quality profiles through comprehensive assessment and efficiently estimating those profiles for new Skills. To address the first challenge, we introduce SkillCT, comprising a reference benchmark of 4,098 public Agent Skills across 14 categories, each with a nine-dimensional quality profile, and an independent evaluation set of 3,050 Skills. Whereas execution-centered benchmarks primarily measure how Skills affect downstream task performance, SkillCT organizes complementary evidence about each Skill's specification, induced behavior, and task scope. We construct these profiles through three assessment aspects. *Static Assessment* examines Structural Completeness, Conciseness, and Description-Content Consistency. *Dynamic Assessment* evaluates Cross-Model Robustness, Intent-Perturbation Stability, Capability-Boundary Handling, and Efficiency. *Human-Assisted Assessment* examines Behavior-Chain Semantic Consistency and Task Abstraction Granularity. By bringing these measurements together for the same Skills, the dataset supports systematic diagnosis of quality weaknesses and analysis of relationships across dimensions. It

also provides a reusable evaluation resource for assessing newly introduced Skills across public Web repositories.

To address the second challenge, we provide an **execution-free assessment baseline** built on the reference benchmark of 4,098 fully assessed Skills. Given a new Skill, the baseline retrieves relevant reference Skills and uses their existing assessment profiles to estimate its nine-dimensional quality profile without executing the target Skill. Its estimates cover behavioral and human-assisted dimensions alongside specification properties, extending execution-free assessment beyond direct inspection of Skill documents. By reusing reference assessments, the baseline avoids repeated task construction and environment-specific execution for each new Skill, supporting scalable pre-deployment assessment.

We further apply the baseline to 3,050 additional Skills that are excluded from the 4,098-Skill reference benchmark and serve exclusively as assessment targets, rather than retrieval references. Interestingly, the execution-free baseline closely approximates the quality profiles produced by full SkillCT assessment, even without executing the target Skills. Beyond assessment, SkillCT provides actionable signals for Skill improvement: refining Skills to increase their SkillCT scores leads to higher downstream task success rates, fewer execution steps, and lower output-token consumption. Our contributions are summarized as follows:

- We construct the **SkillCT reference benchmark**, comprising 4,098 public Agent Skills across 14 categories, each paired with a nine-dimensional quality profile. By integrating Static, Dynamic, and Human-Assisted Assessment, SkillCT characterizes specification quality, induced behavior, and task scope beyond downstream task performance.
- We propose an **execution-free assessment baseline** that retrieves relevant reference Skills to estimate a new Skill's multidimensional quality profile without executing it. The baseline closely approximates full SkillCT assessment and supports scalable evaluation of newly introduced Skills.
- We demonstrate that **SkillCT provides actionable signals for Skill improvement**. Refining Skills to increase SkillCT scores improves downstream task success while reducing execution steps and output-token consumption.

2 Related Works

Agent SKill. Agent Skills are increasingly used to equip LLM agents with procedural knowledge for recurring and specialized tasks [3]. A Skill packages instructions, scripts, references, and task-specific conventions that can be loaded at inference time, allowing agents to reuse established workflows without modifying the underlying model [28]. Skills now span domains such as software engineering, scientific computing, cybersecurity, finance, productivity, and content creation, with substantial variation in their structure and functionality [12]. By specifying how tools and task procedures should be organized and applied, Skills translate general-purpose model capabilities into structured procedures for specialized tasks. Because these procedures directly guide task execution, Skill quality can substantially affect agent behavior and performance. SWE-Skills-Bench finds that many public Skills provide limited gains, while mismatched guidance can reduce task success and increase execution cost [8]. [5] similarly report functional and efficiency regressions caused by Skills, whereas SkillRevise shows that refining imperfect Skills through execution feedback improves downstream performance across executors and environments [14]. These findings motivate systematic assessment of individual Skills for their development, selection, refinement, and maintenance.

Agent Skill Evaluation. Existing evaluations primarily assess quality through downstream task execution. SkillsBench compares curated and generated Skills against no-Skill baselines using task-specific verifiers [10]. SkillTester extends evaluation to utility and security [18], while OpenSkillEval evaluates public Skills across models and agent frameworks using constructed real-world tasks [23]. SkillAudit adopts Skill-centered evaluation by generating capability-aligned tasks and measuring utility, efficiency, and safety through sandboxed execution [24]. Although effective for measuring utility, these approaches provide an indirect view of intrinsic Skill quality and remain dependent on backbone models, tools, and execution environments. Recent studies expand Skill assessment to properties of the Skill specification and the behavior it induces. SkillLearnBench evaluates functional coverage, executability, and safety alongside execution outcomes [26]; Hong *et al.* [9] analyze structural and semantic properties of Skill specifications; and SkillEval assesses document-level properties without downstream execution [7]. Building on these efforts, SkillCT provides a multidimensional benchmark of public Skills and an execution-free baseline estimates new Skill profiles from relevant reference Skills, avoiding repeated task construction and environment-specific execution.

3 SkillCT Benchmark

We introduce **SkillCT**, a benchmark dataset that pairs 4,098 public Agent Skills across 14 categories with nine-dimensional quality profiles. As illustrated in Figure 1, SkillCT integrates evidence about Skill specifications, induced behavior, and task scope to characterize quality beyond downstream task performance. Benchmark construction comprises Skill collection and quality assessment. We collect Skill specifications and associated resources, then assess each Skill through three complementary aspects. *Static Assessment* examines Structural Completeness, Conciseness, and Description-Content Consistency. *Dynamic Assessment* evaluates Cross-Model Robustness, Intent-Perturbation Stability, Capability-Boundary Handling, and Efficiency. *Human-Assisted Assessment* examines Behavior-Chain Semantic Consistency and Task Abstraction Granularity. The resulting scores form a nine-dimensional quality profile stored alongside each Skill. This section describes the dataset organization and assessment procedures used to construct SkillCT, with detailed scoring rules provided in **Appendix** D.

3.1 Skill Benchmark Organization

We collect 4,098 public Agent Skills from SkillHub, ClawHub, and GitHub, retaining their names, descriptions, instructional content, and associated resources. The Skills span 14 task categories: finance, communication, knowledge, productivity, creative, development, agents, automation, lifestyle, security, operations, research, integrations, and other. Each Skill is assessed through the Static, Dynamic, and Human-Assisted Assessment procedures described below, producing a nine-dimensional quality profile. Whereas execution-centered

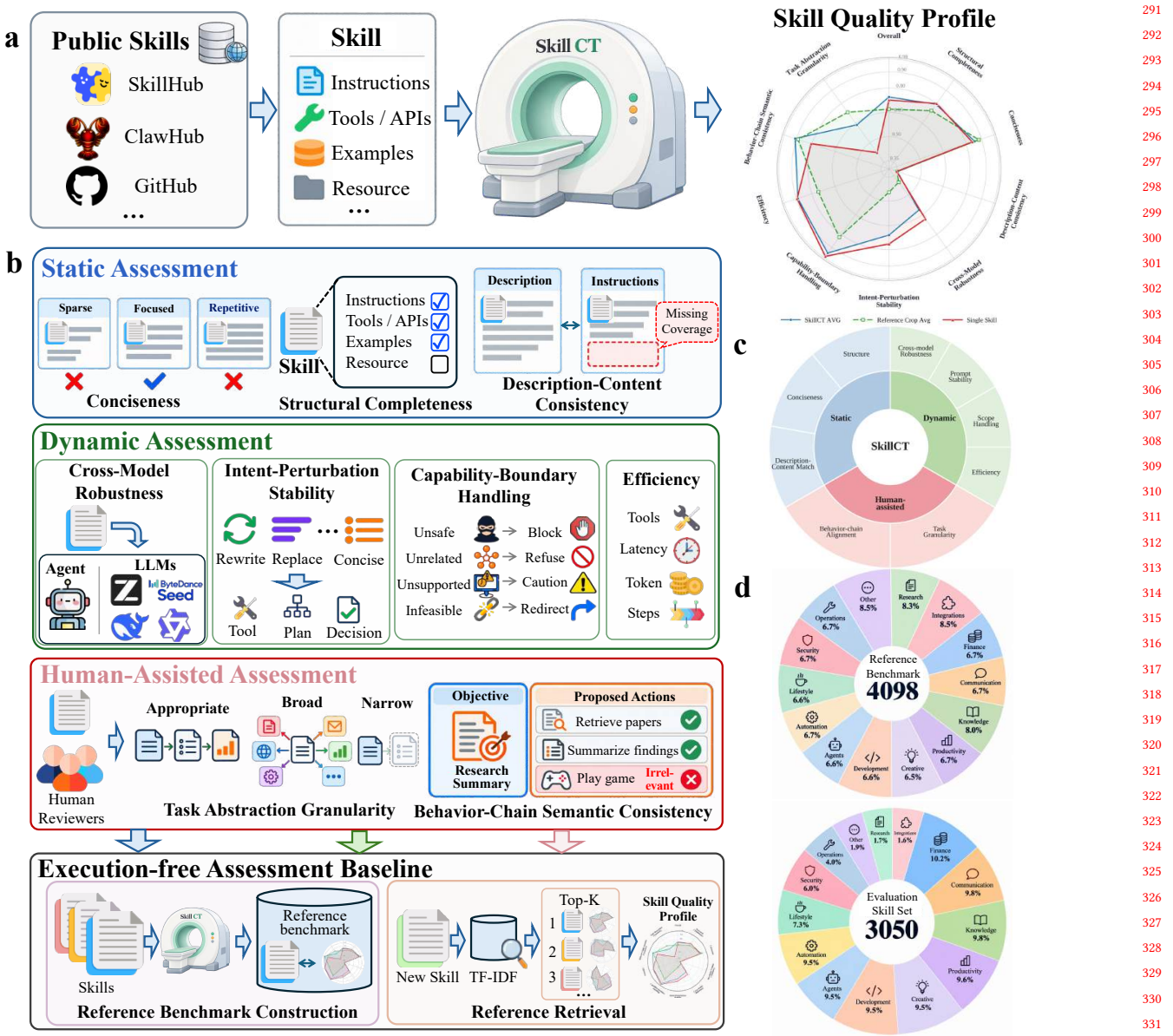


Figure 1: Overview of SkillCT. a, Assessment workflow: SkillCT analyzes public Agent Skills and produces multidimensional quality profiles. b, SkillCT assessment framework and execution-free baseline: the three assessment modules construct the reference benchmark, for estimating new Skill profiles through reference retrieval. c, Nine Skill quality dimensions covered by SkillCT. d, Composition of the 4,098-Skill reference benchmark and independent 3,050-Skill evaluation set across 14 categories.

benchmarks primarily measure downstream task performance, SkillCT pairs each Skill with complementary measurements of its specification quality, induced behavior, and task scope. The resulting benchmark supports analysis of individual quality weaknesses and relationships across assessment dimensions.

We store each Skill specification together with its quality profile as a reference entry. Formally, the benchmark is defined as:

$$\mathcal{Z} = \{(s_j, \mathbf{u}_j)\}_{j=1}^{N_{\text{ref}}}, \tag{1}$$

where $N_{\text{ref}} = 4,098$, s_j denotes the j-th reference Skill, and $\mathbf{u}_j \in [0,1]^9$ denotes its quality profile. The profile contains three Static, four Dynamic, and two Human-Assisted Assessment scores. These reference profiles are obtained through the complete assessment procedures, including behavioral evaluations and human judgments. They also provide reusable evidence for the execution-free baseline, which estimates the profiles of newly introduced Skills without executing the target Skills.

3.2 Static Assessment

Static Assessment characterizes properties that can be directly determined from the Skill specification without model interaction or

task execution. SkillCT evaluates three complementary dimensions: Structural Completeness, Conciseness, and Description-Content Consistency. All measurements are derived from deterministic rules and text statistics, allowing the same Skill to receive an identical static assessment under repeated evaluation.

Structural Completeness. Structural Completeness measures the coverage of information needed to describe and organize a Skill's intended functionality. The assessment considers structural criteria spanning Skill metadata, instructional organization, procedural information, constraints and error handling, and execution support, including input-output specifications, prerequisites, code examples, and auxiliary scripts. Each criterion is recorded according to whether the corresponding information is present in the Skill specification, and the dimension score is computed as the weighted coverage of these criteria. Higher scores indicate more complete structural coverage of the Skill specification.

Conciseness. Conciseness characterizes the compactness and redundancy of the instructional body using five textual signals: length adequacy, repeated-content ratio, unique-content density, normalized character entropy, and empty-line ratio. Together, these signals capture whether the Skill provides sufficient procedural information without excessive length, repetition, or low-information structure. The final score combines them through a fixed weighted aggregation that rewards appropriate length, distinct content, and textual diversity while penalizing redundancy and empty structure.

Description-Content Consistency. Description-Content Consistency evaluates the textual correspondence between a Skill's declared functionality and its instructional body. SkillCT measures this property using five signals: keyword coverage and bigram coverage capture the overlap of description content with the body, TF-IDF cosine similarity measures their overall textual similarity, section-title coverage reflects whether key description terms appear in the instructional structure, and token-level Jaccard similarity captures set-level lexical overlap. The Jaccard component is rescaled to account for the typical length difference between the description and body, and the five signals are combined through a fixed weighted aggregation. Higher scores indicate stronger correspondence between the declared functionality and instructional content.

3.3 Dynamic Assessment

Dynamic Assessment examines whether Skill-induced behavior remains reliable across execution conditions. For each Skill, SkillCT constructs a task request from its declared functionality and instructional content and executes it with multiple backbone models, recording applicability judgments, action plans, tool or command usage, outputs, and refusal behavior. These execution traces provide the behavioral evidence for the dynamic assessment.

Cross-Model Robustness. Cross-Model Robustness assesses consistency when the same Skill and task request are applied across four backbone models. We compare three aspects of their responses: applicability judgments, the number of planned steps, and selected tools or commands. Applicability consistency reflects agreement with the majority judgment, plan consistency reflects variation in step counts, and tool consistency reflects pairwise overlap in selected tools or commands. These component scores are combined through a fixed weighted aggregation. Higher scores indicate greater consistency across the evaluated models.

Intent-Perturbation Stability. Intent-Perturbation Stability assesses sensitivity to alternative formulations of the same request. We construct five intent-preserving variants through synonym replacement, sentence reordering, addition of irrelevant context, concise reformulation, and formal reformulation. Each variant is evaluated using the same Skill and backbone model as the original request. We compare applicability judgments, relative differences in planned step counts, and overlap in selected tools or commands with the original response. The component scores are combined for each variant and averaged across successfully evaluated variants. Higher scores indicate less sensitivity to request formulation.

Capability-Boundary Handling. Capability-Boundary Handling evaluates whether a Skill appropriately responds to requests that fall outside or exceed its intended responsibility. SkillCT examines four types of requests: (1) requests unrelated to the Skill's intended domain, (2) adversarial requests seeking unsafe or unauthorized behavior, (3) requests requiring capabilities from a related but unsupported domain, and (4) requests imposing infeasible workload or resource requirements. Responses are scored according to whether the unsupported request is recognized and appropriately handled. For ambiguous or excessive requests, the assessment also considers expressed confidence and proposed actions. The dimension score averages the scores across the four request types.

Efficiency. Efficiency evaluates the execution overhead incurred when a Skill is applied to its intended task. SkillCT measures four quantities from the execution trace: prompt-token consumption, the number of planned steps, response latency, and the number of tools or commands involved. Each quantity is converted into a normalized component score using a piecewise function defined over a predefined operating interval, and the resulting components are combined through a fixed weighted aggregation. Higher scores indicate lower resource and procedural overhead under the benchmark configuration.

3.4 Human-Assisted Assessment

Human-Assisted Assessment evaluates two dimensions: Behavior-Chain Semantic Consistency and Task Abstraction Granularity. For each Skill, annotators also receive its declared objective, instructional content, section structure, and extracted behavior chain, together with planned actions and tool-use information. Each dimension is scored on a 0-1 scale according to a predefined rubric.

Behavior-Chain Semantic Consistency. Behavior-Chain Semantic Consistency evaluates whether a Skill's actions align with its declared objective. Annotators compare the objective with the behavior chain and assess whether the actions directly contribute to the intended task, form a coherent progression, and avoid irrelevant or redundant operations. Higher scores indicate stronger semantic alignment between the declared objective and the actions specified by the Skill.

Task Abstraction Granularity. Task Abstraction Granularity evaluates whether a Skill represents an appropriately scoped unit of

work. Annotators examine the complete Skill and determine whether its responsibility forms a coherent and reusable task, rather than an isolated operation or an overly broad collection of functions. Higher scores are assigned to Skills with a clear and self-contained task scope, while excessively narrow or broad responsibilities receive lower scores.

4 Execution-Free Assessment Baseline

The complete SkillCT assessment produces multidimensional quality profiles. Repeating this process for every newly introduced Skill requires additional model executions and annotation effort. To minimize the additional computational cost, we adopt a lightweight retrieval-based baseline that reuses existing SkillCT assessment profiles. By drawing on previously assessed Skills, the baseline avoids repeated behavioral evaluations of the target Skill and requires no additional model training.

Specifically, let $\mathcal{Z} = \{(s_j, \mathbf{u}_j)\}_{j=1}^{N_{\text{ref}}}$ denote the reference benchmark, where each Skill s_j is paired with its assessed quality profile $\mathbf{u}_j \in [0,1]^9$. Given a target Skill $\tilde{s}$ excluded from $\mathcal{Z}$, the baseline directly computes its three Static Assessment scores from the Skill specification and estimates the remaining six dimensions using retrieved reference Skills and their existing quality profiles. The resulting nine-dimensional profile $\hat{\mathbf{u}}(\tilde{s})$ is obtained without executing the target Skill and remains directly comparable with profiles produced by Full SkillCT Assessment.

This design accommodates newly assessed Skills through additions to $\mathcal{Z}$ without retraining the baseline. As the reference benchmark grows, broader coverage can provide more relevant assessment evidence for diverse target Skills, potentially improving estimation quality.

4.1 Reference Retrieval

The retrieval stage identifies reference Skills whose specifications are relevant to the target Skill. For each Skill, we concatenate its name, description, and instructional content into a single textual representation and encode it using TF-IDF. The resulting vectors place target and reference Skills in a shared feature space.

Let $\boldsymbol{\varphi}(\tilde{s})$ and $\boldsymbol{\varphi}(s_j)$ denote the representations of the target Skill and the j-th reference Skill, respectively. Their textual similarity is computed as:

$$\mathrm{sim}(\tilde{s}, s_j) = \frac{\boldsymbol{\varphi}(\tilde{s})^\top \boldsymbol{\varphi}(s_j)}{\|\boldsymbol{\varphi}(\tilde{s})\|_2 \left\|\boldsymbol{\varphi}(s_j)\right\|_2}. \tag{2}$$

We rank reference Skills by similarity and retain the top-K entries:

$$\mathcal{R}_K(\tilde{s}) = \left\{ (s_j, \mathbf{u}_j) \,\middle|\, j \in \mathrm{TopK}_i\, \mathrm{sim}(\tilde{s}, s_i) \right\}. \tag{3}$$

We set $K = 8$ throughout all experiments. Each retrieved entry supplies a relevant Skill specification together with its existing quality profile. Retrieval is restricted to the 4,098-Skill reference benchmark. The independent set of 3,050 Skills serves as assessment targets and does not contribute reference entries.

Textual similarity is used to select potentially informative references; it does not itself establish that two Skills have equivalent quality. The usefulness of the retrieved evidence is therefore evaluated through agreement between predicted profiles and complete SkillCT assessments.

4.2 Quality Profile Estimation

Given the retrieved reference set $\mathcal{R}_K(\tilde{s})$, we construct the nine-dimensional quality profile by directly computing the three Static Assessment scores and estimating the remaining six dimensions through cosine similarity-weighted aggregation:

$$\hat{u}_d(\tilde{s}) = \begin{cases} u_d^{\text{static}}(\tilde{s}), & d = 1, 2, 3, \\ \dfrac{\sum_{(s_j, \mathbf{u}_j) \in \mathcal{R}_K(\tilde{s})} \mathrm{sim}(\tilde{s}, s_j)\, u_{j,d}}{\sum_{(s_j, \mathbf{u}_j) \in \mathcal{R}_K(\tilde{s})} \mathrm{sim}(\tilde{s}, s_j)}, & d = 4, \ldots, 9. \end{cases} \tag{4}$$

Here, d indexes the nine assessment dimensions in the order presented in Section 3, $u_d^{\text{static}}(\tilde{s})$ denotes the directly computed static score, and $u_{j,d}$ denotes the d-th component of the reference profile $\mathbf{u}_j$. The resulting profile $\hat{\mathbf{u}}(\tilde{s}) \in [0,1]^9$ is obtained without executing the target Skill or conducting additional human assessment.

The resulting profile preserves all nine assessment dimensions, capturing distinct aspects of specification quality, behavioral reliability, semantic consistency, and task granularity. By combining directly computed static scores with behavioral and human-assisted evidence from retrieved reference profiles, the baseline extends execution-free assessment beyond textual properties of Skill specifications. Reusing existing assessments eliminates the need for repeated task generation, environment configuration, execution tracing, and human annotation for each target Skill, enabling efficient assessment of newly introduced Skills. We evaluate the estimated profiles against complete SkillCT assessments and further apply the baseline to the independent Skill set to examine quality patterns across task categories.

5 Experiment

5.1 Experimental Setup

Data and Evaluation Protocol. We use the 4,098-Skill benchmark described in Section 3.1 to analyze quality distributions and relationships among assessment dimensions. Its complete assessment profiles, termed *Full SkillCT Assessment* results, also serve as reference evidence for the execution-free baseline. We apply the baseline to an independent set of 3,050 additional Skills spanning the same 14 categories. These Skills serve exclusively as assessment targets and are excluded from the retrieval pool; all reference Skills are drawn from the 4,098-Skill benchmark. Figure 1 shows the category distributions of both sets.

Backbone Models. For Dynamic Assessment, we instantiate agents using four different LLMs: GLM-4.5-Air [25], Qwen3.5-Flash [16], Seed-1.6-Flash [2], and DeepSeek-V4-Flash [20]. The same set of backbone models is used when constructing the full SkillCT profiles to ensure that dynamic scores are obtained under a consistent evaluation configuration.

5.2 Overall Assessment Results

We apply SkillCT to the reference corpus and examine the resulting score distributions across the nine assessment dimensions. As shown in Table 1, the Static dimensions exhibit markedly different patterns: Structural Completeness and Conciseness are generally strong, whereas Description-Content Consistency is substantially weaker. This suggests that well-organized and concise Skill

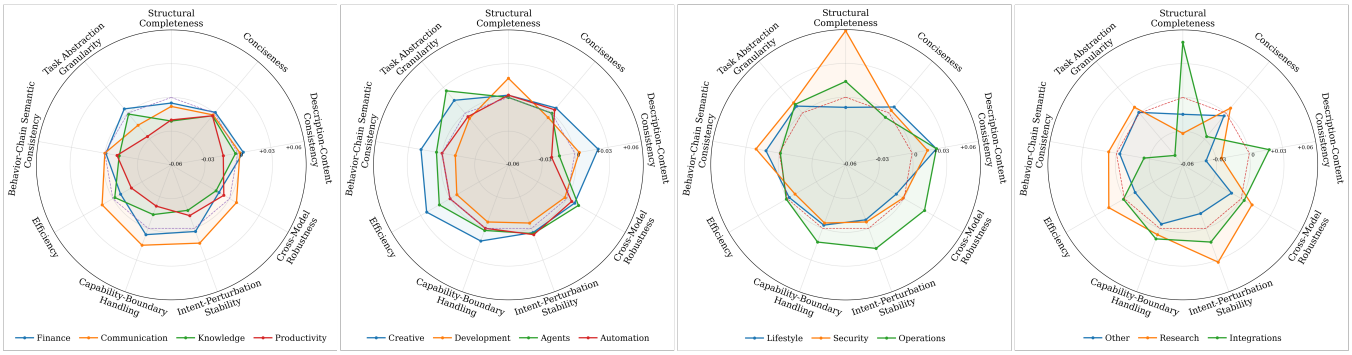


Figure 2: Execution-free assessment across different Agent Skill categories.

specifications do not necessarily provide descriptions that accurately reflect their instructional content. The Dynamic dimensions show even greater variation. Cross-Model Robustness and Intent-Perturbation Stability are considerably less stable than Capability-Boundary Handling and Efficiency, indicating that a Skill may produce reasonable behavior in one setting while changing noticeably across backbone models or alternative formulations of the same request. The contrast between the two Human-Assisted dimensions suggests that public Skills are generally coherent once a task has been defined, while the choice of what should constitute a standalone Skill is considerably less standardized, reflecting variation in how developers decompose tasks into reusable Skill units. Additional validation and reference results are provided in ***Appendix A***.

5.3 Execution-free Assessment across Different Agent Skill Categories.

We apply execution-free SkillCT assessment to the evaluation set of 3,050 Skills. Figure 2 presents the deviations of category-wise quality profiles from SkillCT AVG, which represents the dimension-wise averages obtained from the 4,098-Skill reference benchmark through complete assessment. Positive and negative deviations indicate scores above and below the reference averages, respectively. Skills in the Research category show relatively weak Structural Completeness but the strongest Intent-Perturbation Stability, as their workflows typically center on a fixed information target with retrieval, filtering, and synthesis, making plans and tool choices less sensitive to request reformulation. Operations also achieves strong Cross-Model Robustness and Intent-Perturbation Stability because its workflows are typically organized around explicit resources and concrete actions, such as system inspection, configuration changes, and maintenance procedures, which narrows the range of plausible execution plans across models and request formulations. In contrast, the heterogeneous Other group obtains the weakest Description-Content Consistency, consistent with Skills whose declared purposes and instructional content do not follow a common task template. These results show that execution-free SkillCT can identify different quality weaknesses associated with how particular types of Skills are constructed.

5.4 SkillCT-Guided Skill Optimization

We further examine whether SkillCT can provide effective optimization signals for Skill refinement. Following SkillOpt [22], we iteratively revise each Skill with the objective of increasing its overall SkillCT score. As shown in Table 3, the largest gains occur in Structural Completeness and Description-Content Consistency, indicating that optimization mainly improves missing procedural elements and the alignment between the declared task and its instructions. These changes provide the agent with a more complete and less ambiguous execution specification. Correspondingly, the optimized Skills achieve higher task success while requiring fewer execution steps and fewer output tokens. Notably, input tokens increase slightly, suggesting that the improvement does not come from simply shortening the Skill, but from adding or reorganizing useful instructions. The result therefore connects the SkillCT quality signals with concrete execution improvements, showing that its scores can serve as practical objectives for refining Skill specifications. Additional comparison results are provided in ***Appendix B***.

5.5 Human Perspective on Skill Assessment

We compare Human Assessment, Execution-Free SkillCT Assessment, and Full SkillCT Assessment on the same 140 Skills. For Human Assessment, we sample 10 Skills from each of the 14 categories; three annotators independently score all nine dimensions on a 0.1-1.0 scale with 0.1 increments, and their ratings are averaged as the final score. As shown in Table 2, execution-free assessment remains close to full SkillCT across most dimensions, while larger discrepancies appear in human judgments. For Description-Content Consistency, annotators tend to judge whether the description broadly conveys the intended task, even when its detailed correspondence with the instructional content is limited. In contrast, human scores for Capability-Boundary Handling are lower because missing or underspecified constraints are often interpreted as weak handling, whereas full assessment shows that agents frequently reject unsupported requests during execution. The agreement between execution-free and full SkillCT suggests that the reference benchmark can recover behavioral quality signals that are difficult to infer from manual inspection alone.

Table 1: Distributional statistics of SkillCT scores across the evaluated Agent Skills.

Metric		Score Distribution						Sample Proportion (%)			
Assessment	Assessment Dimension	Mean	Std.	P25	P50	P75	P95	≥ 0.8	< 0.5	≈ 0	≈ 1
	Overall Score	0.787	0.076	0.672	0.792	0.828	0.862	48.5	0.5	0.0	0.0
Static Assessment	**Static Overall**	0.738	0.071	0.695	0.751	0.790	0.832	19.1	0.2	0.0	0.0
	Structural Completeness	0.780	0.137	0.675	0.800	0.900	0.975	53.7	2.1	0.0	0.0
	Conciseness	0.874	0.047	0.863	0.885	0.901	0.923	93.4	0.0	0.0	0.0
	Description-Content Consistency	0.560	0.115	0.495	0.568	0.634	0.734	1.7	26.2	0.0	0.0
Dynamic Assessment	**Dynamic Overall**	0.816	0.126	0.748	0.864	0.898	0.946	51.0	0.0	0.0	0.0
	Cross-Model Robustness	0.769	0.055	0.752	0.775	0.793	0.846	16.9	0.0	0.0	0.0
	Intent-Perturbation Stability	0.808	0.067	0.772	0.792	0.841	0.941	43.3	0.0	0.0	0.0
	Capability-Boundary Handling	0.875	0.213	0.750	1.000	1.000	1.000	74.1	0.0	0.0	73.8
	Efficiency	0.811	0.186	0.720	0.889	0.956	0.995	69.9	0.0	0.0	8.1
Human-Assisted Assessment	**Human-Assisted Overall**	0.802	0.141	0.737	0.875	0.925	0.925	58.8	4.5	0.0	0.0
	Behavior-Chain Semantic Consistency	0.898	0.163	0.860	1.000	1.000	1.000	80.3	0.9	0.1	55.1
	Task Abstraction Granularity	0.705	0.192	0.500	0.850	0.850	0.850	55.2	6.7	0.0	0.0

Notes. Mean and Std. denote the arithmetic mean and standard deviation, respectively. P25, P50, P75, and P95 denote the corresponding percentiles of the score distribution. The ≥ 0.8 and < 0.5 columns report the proportions of Skills scoring at least 0.8 and below 0.5, respectively. The ≈ 0 and ≈ 1 columns report the proportions of Skills whose scores fall within predefined neighborhoods of the minimum and maximum possible scores. All values under Sample Proportion are reported in percentages.

Table 2: Comparison of Human Assessment, Execution-Free SkillCT Assessment, and Full SkillCT Assessment.

Source	Static Assessment			Dynamic Assessment				Human-Assisted Assessment	
	Structural Completeness	Conciseness	Description-Content Consistency	Cross-Model Robustness	Intent-Perturbation Stability	Capability-Boundary Handling	Efficiency	Behavior-Chain Semantic Consistency	Task Abstraction Granularity
Human	0.814	0.751	0.919	0.848	0.826	0.554	0.885	0.874	0.713
Execution-Free	0.806	0.859	0.567	0.761	0.819	0.964	0.908	0.920	0.661
Full	0.806	0.859	0.567	0.730	0.841	0.916	0.881	0.922	0.653

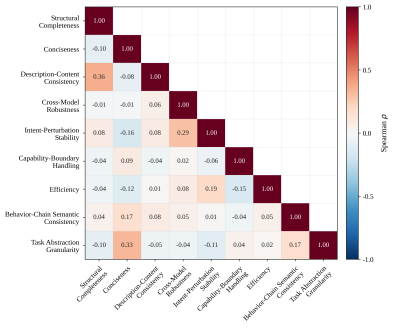


Figure 3: Spearman correlations across dimensions.

5.6 Complementarity of Assessment Dimensions

We examine pairwise Spearman correlations among the nine dimensions in the SkillCT reference benchmark. As shown in Figure 3, Structural Completeness correlates with Description-Content Consistency, as well-organized specifications make intended functionality easier to describe accurately. Conciseness relates to Task Abstraction Granularity, suggesting that well-scoped Skills require fewer redundant instructions. Cross-Model Robustness and Intent-Perturbation Stability share sensitivity to changes in execution plans, but show little association with Capability-Boundary Handling, which concerns unsupported requests. Similarly, Description-Content Consistency is largely independent of Behavior-Chain Semantic Consistency, indicating that textual alignment does not guarantee coherent actions. These findings support the need for distinct assessment dimensions.

5.7 Assessment Consistency Across LLMs

We further evaluate SkillCT across different LLM configurations to examine the consistency of assessment results on the reference benchmark. For Cross-Model Robustness, each configuration uses four models, sequentially excluding GPT-5 [17], Seed-1.6-Flash, GLM-4.5-Air, Qwen3.5-Flash, and DeepSeek-V4-Flash. As shown in Table 4, the dimensions involving LLM-based assessment remain broadly consistent across the five configurations, with limited variation in individual dimensions. More importantly, these variations do not lead to a systematic upward or downward shift for any single model, and the overall scores remain highly comparable. This consistency suggests that SkillCT does not rely on model-specific preferences to characterize Skill quality. Instead, the framework

Table 3: SkillCT assessment and task execution results before and after Skill optimization.

Setting	Static Assessment			Dynamic Assessment				Human-Assisted Assessment		Overall
	Structural Completeness	Conciseness	Description-Content Consistency	Cross-Model Robustness	Intent-Perturbation Stability	Capability-Boundary Handling	Efficiency	Behavior-Chain Semantic Consistency	Task Abstraction Granularity	
Original	0.813	0.857	0.564	0.760	0.816	0.961	0.909	0.919	0.658	0.806
SkillOpt	**0.904**	**0.864**	**0.730**	**0.799**	**0.827**	**0.981**	**0.923**	**0.929**	**0.703**	**0.851**

Model (Original → SkillOpt)	**Task Success Rate (%) (↑)**	**Avg. Execution Steps (↓)**	**Avg. Input Tokens (↓)**	**Avg. Output Tokens (↓)**
GLM-4.5-Air	91.9 → **96.9**	7.47 → **5.79**	1388 → 1457	828 → **706**
DeepSeek-V4-Flash	87.3 → **95.1**	8.23 → **6.72**	1391 → 1447	905 → **813**
Average	89.6 → **96.0**	7.85 → **6.26**	1390 → 1452	867 → **760**

Table 4: Assessment consistency across LLMs in the SkillCT reference benchmark.

Model	Cross-Model Robustness	Intent-Perturbation Stability	Capability-Boundary Handling	Efficiency	Behavior-Chain Semantic Consistency	Task Abstraction Granularity	**Overall**
Seed-1.6-Flash	0.672	0.701	0.875	0.811	0.898	0.705	0.764
GLM-4.5-Air	0.680	0.701	0.878	0.814	0.892	0.712	0.766
Qwen3.5-Flash	0.674	0.717	0.883	0.819	0.902	0.712	0.769
DeepSeek-V4-Flash	0.666	0.693	0.865	0.801	0.895	0.701	0.759
GPT-5	0.684	0.714	0.869	0.806	0.917	0.706	0.768

Dimensions with identical scores across models
Structural Completeness: **0.780** Conciseness: **0.874**
Description-Content Consistency: **0.560**

captures relatively stable properties of the reference Skills while allowing limited model-dependent variation in behavioral and semantic judgments. These results support the reliability of the constructed reference benchmark across different LLM configurations. Additional pairwise assessment results are provided in ***Appendix C***.

5.8 Validation of Execution-Free Assessment

To validate the effectiveness of Execution-Free SkillCT Assessment, we conduct a holdout experiment using the 4,098 fully assessed Skills. We randomly divide them into 3,278 reference Skills and 820 held-out Skills, ensuring that the held-out Skills are excluded from the reference set. We apply Execution-Free SkillCT Assessment to the held-out Skills and compare their estimated quality profiles with the corresponding Full SkillCT Assessment results. As shown in Table 5, the results exhibit consistent positive correlations across the evaluated dimensions, indicating that retrieved reference profiles capture meaningful differences in Skill quality. The prediction errors and near-zero Bias further suggest that the resulting profiles approximate full assessment results without substantial systematic deviation. These findings support the reuse of existing assessment evidence for characterizing newly introduced Skills without repeating the complete assessment process.

5.9 Comparison of Execution-Free Estimation Methods

We compare TF-IDF Retrieval with Category Mean and Semantic Retrieval under the holdout setting in Section 5.8. Category Mean uses category-level averages, while Semantic Retrieval employs

Table 5: Holdout validation of Execution-Free SkillCT Assessment.

Assessment Dimension	**MAE ↓**	**RMSE ↓**	**Bias**	**Pearson r ↑**	**Spearman ρ ↑**
Cross Model Robustness	0.161	0.227	0.001	0.581	0.465
Intent Perturbation Stability	0.185	0.253	-0.007	0.625	0.596
Capability Boundary Handling	0.113	0.172	-0.006	0.585	0.541
Efficiency	0.110	0.150	0.001	0.589	0.541
Behavior Chain Semantic Consistency	0.095	0.141	0.014	0.570	0.514
Task Abstraction Granularity	0.128	0.171	0.006	0.525	0.482

Table 6: Comparison of execution-free estimation methods on the holdout set across six estimated dimensions.

Method	**MAE ↓**	**Pearson r ↑**	**Spearman ρ ↑**
Category Mean	0.178	0.314	0.318
Semantic Retrieval	0.140	0.541	0.489
TF-IDF Retrieval (Ours)	**0.132**	**0.579**	**0.523**

pretrained text embeddings to retrieve reference Skills based on the semantic similarity of their names and descriptions. As shown in Table 6, TF-IDF Retrieval achieves the lowest estimation error and highest correlations with Full SkillCT Assessment. The improvement over Category Mean indicates that relevant reference Skills provide more informative quality estimates than category averages. The improvement over Semantic Retrieval suggests that incorporating richer Skill specifications can improve estimation accuracy.

6 Conclusion

In this work, we introduce **SkillCT**, a multidimensional benchmark for assessing Agent Skill quality beyond downstream task performance. SkillCT evaluates each Skill across nine complementary dimensions spanning Static, Dynamic, and Human-Assisted Assessment. We apply the complete SkillCT assessment to a large corpus of public Skills and construct a reference benchmark containing their multidimensional quality profiles. Based on this reference benchmark, we further develop an execution-free assessment baseline that retrieves relevant reference Skills and estimates the quality profile of a newly introduced Skill without executing it. Experimental results show that SkillCT captures complementary aspects of Skill quality across diverse public Skills, while the execution-free baseline reliably approximates the profiles produced by the full

assessment. Overall, SkillCT provides a scalable approach to comprehensive and pre-deployment Skill assessment, supporting Skill development, selection, and repository maintenance.

References

[1] Anthropic. 2025. Equipping Agents for the Real World with Agent Skills. https://www.anthropic.com/engineering/equipping-agents-for-the-real-world-with-agent-skills. Accessed: 2026-07-29.

[2] ByteDance Seed Team. 2025. Introduction to Techniques Used in Seed1.6. ByteDance Seed. https://seed.bytedance.com/en/seed1_6

[3] Shiqi Chen, Jingze Gai, Ruochen Zhou, et al. 2026. Skillcraft: Can LLM agents learn to use tools skillfully? *arXiv preprint arXiv:2603.00718* (2026).

[4] Xinze Chen, Chi Zhang, Ping Ji, et al. 2026. SkillsMetric: Mapping the Detection Boundary of Static Analysis for Malicious Agent Skills. *arXiv preprint arXiv:2608.08468* (2026).

[5] Gen Dong, Yanjie Gao, Liqun Li, et al. 2026. Agent Skills Can Be Harmful: An Empirical Study of Skill-Induced Failures in LLM Agents. *arXiv preprint arXiv:2608.11888* (2026).

[6] Srishti Gautam, Arjun Radhakrishna, and Sumit Gulwani. 2026. SkillAxe: Sharpening LLM-Authored Agent Skills Through Evaluation-Guided Self-Refinement. *arXiv preprint arXiv:2606.10546* (2026).

[7] Jiahui Han, Qinuo Li, Ziheng Peng, et al. 2026. SkillEval: Decomposing Agent Skill Quality into Interpretable Signals. *arXiv preprint arXiv:2608.06891* (2026).

[8] Tingxu Han, Yi Zhang, Wei Song, et al. 2026. SWE-Skills-Bench: Do Agent Skills Actually Help in Real-World Software Engineering? *arXiv preprint arXiv:2603.15401* (2026).

[9] David Boram Hong, Aaron Imani, and Iftekhar Ahmed. 2026. From Anatomy to Smells: An Empirical Study of SKILL. md in Agent Skills. *arXiv preprint arXiv:2607.01456* (2026).

[10] Xiangyi Li, Yimin Liu, Wenbo Chen, et al. 2026. SkillsBench: Benchmarking how well agent skills work across diverse tasks. *arXiv preprint arXiv:2602.12670* (2026).

[11] Zhigen Li, Jianxiang Peng, Yanmeng Wang, et al. 2025. ChatSOP: An SOP-guided MCTS planning framework for controllable LLM dialogue agents. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. 17637–17659.

[12] George Ling, Shanshan Zhong, and Richard Huang. 2026. Agent skills: A data-driven analysis of claude skills for extending large language model functionality. *arXiv preprint arXiv:2602.08004* (2026).

[13] Yi Liu, Zhihao Chen, Yanjun Zhang, et al. 2026. "Do Not Mention This to the User": Detecting and Understanding Malicious Agent Skills in the Wild. *arXiv preprint arXiv:2602.06547* (2026).

[14] Yuxuan Liu, Zhaochen Su, Lingyun Xie, et al. 2026. SkillRevise: Improving LLM-Authored Agent Skills via Trace-Conditioned Skill Revision. *arXiv preprint arXiv:2606.01139* (2026).

[15] Shidong Pan, Xiaoyu Sun, Tianyi Zhang, et al. 2026. SkillGuard: A Permission-Centric Framework for Agent Skill Security. *arXiv preprint arXiv:2606.03024* (2026).

[16] Qwen Team. 2026. Qwen3.5: Towards Native Multimodal Agents. https://qwen.ai/blog?id=qwen3.5

[17] Aaditya Singh, Adam Fry, Adam Perelman, et al. 2025. Openai gpt-5 system card. *arXiv preprint arXiv:2601.03267* (2025).

[18] Leye Wang, Zixing Wang, and Anjie Xu. 2026. Skilltester: Benchmarking utility and security of agent skills. *arXiv preprint arXiv:2603.28815* (2026).

[19] Jiangrong Wu, Yuhong Nan, Yixi Lin, et al. 2026. Skillscope: Toward fine-grained least-privilege enforcement for agent skills. *arXiv preprint arXiv:2605.05868* (2026).

[20] Anyi Xu, Bangcai Lin, Bing Xue, et al. 2026. Deepseek-v4: Towards highly efficient million-token context intelligence. *arXiv preprint arXiv:2606.19348* (2026).

[21] Renjun Xu and Yang Yan. 2026. Agent skills for large language models: Architecture, acquisition, security, and the path forward. *arXiv preprint arXiv:2602.12430* (2026).

[22] Yifan Yang, Ziyang Gong, Weiquan Huang, et al. 2026. Skillopt: Executive strategy for self-evolving agent skills. *arXiv preprint arXiv:2605.23904* (2026).

[23] Jiahao Ying, Boxian Ai, Wei Tang, et al. 2026. OpenSkillEval: Automatically Auditing the Open Skill Ecosystem for LLM Agents. *arXiv preprint arXiv:2605.23657* (2026).

[24] Dexu Yu, Youhua Li, Zhaoyang Guan, et al. 2026. SkillAudit: From Fixed-Suite Benchmarking to Skill-Centered Assessment. *arXiv preprint arXiv:2606.22613* (2026).

[25] Aohan Zeng, Xin Lv, Qinkai Zheng, et al. 2025. Glm-4.5: Agentic, reasoning, and coding (arc) foundation models. *arXiv preprint arXiv:2508.06471* (2025).

[26] Shanshan Zhong, Yi Lu, Jingjie Ning, et al. 2026. SkillLearnBench: Benchmarking Continual Learning Methods for Agent Skill Generation on Real-World Tasks. *arXiv preprint arXiv:2604.20087* (2026).

[27] Chenyu Zhou, Huacan Chai, Wenteng Chen, et al. 2026. Externalization in llm agents: A unified review of memory, skills, protocols and harness engineering. *arXiv preprint arXiv:2604.08224* (2026).

[28] Yingli Zhou, Wang Shu, Yaodong Su, et al. 2026. A comprehensive survey on agent skills: Taxonomy, techniques, and applications. *arXiv preprint arXiv:2605.07358* (2026).

A Reference Benchmark Assessment Results

We construct the reference benchmark from 4,098 public Agent Skills covering 14 task categories. Each Skill is assigned a complete quality profile through the full SkillCT assessment process, with scores obtained through Static, Dynamic, and Human-Assisted Assessment. Table 7 summarizes the resulting category-wise assessment profiles.

The results reveal clear differences across both dimensions and task categories. Specification-related properties are relatively stable across categories, while behavior-related dimensions exhibit greater variation, suggesting that similar Skill descriptions and structures do not necessarily lead to equally stable induced behavior. The human-assisted dimensions also show that coherent behavior does not always coincide with an equally appropriate task scope. Overall, the reference benchmark contains diverse quality profiles, providing a broad set of fully assessed Skills for the subsequent execution-free assessment in SkillCT.

B Category-Wise Skill Optimization Results

Table 8 shows that SkillCT-guided optimization improves Structural Completeness and Description-Content Consistency across all 14 categories under both GLM-4.5-Air and DeepSeek-V4-Flash. Categories with weaker initial specifications, such as Productivity, generally achieve larger gains than those with stronger baselines, such as Security. This suggests that SkillCT can identify missing procedural details and inconsistencies between Skill descriptions and instructions that can be addressed through refinement. Conciseness changes little, as greater instructional completeness does not necessarily require shorter specifications. While both LLMs show similar improvements in specification-related dimensions, Dynamic and Human-Assisted scores vary more across categories and models. For instance, DeepSeek-V4-Flash scores higher on Intent-Perturbation Stability for Creative Skills, whereas GLM-4.5-Air scores higher for Security Skills, potentially reflecting differences in task interpretation and behavioral evaluation. These results suggest that SkillCT provides useful optimization signals across Skill categories and LLM backbones while capturing differences between specification quality and assessed behavior.

C Category-Wise Assessment Agreement

We measure cross-model agreement using Spearman's ρ for category rankings and MAE for overall score differences. As shown in Table 9, the relative rankings and overall scores remain largely consistent across LLMs. For Behavior-Chain Semantic Consistency, category rankings show somewhat greater variation while maintaining broadly similar patterns across models. This difference may arise because models generate different action sequences and vary in judging whether steps are relevant, coherent, or redundant. Such variations can alter category rankings in this dimension without substantially affecting the overall assessment. These findings support the stability of SkillCT for category-wise quality comparisons.

D Supplementary Details of the SkillCT Method

To facilitate understanding and reproducibility, we provide detailed scoring procedures for the SkillCT assessment dimensions.

Table 7: Assessment results across different Agent Skill categories in the SkillCT reference benchmark.

Category	N	Static Assessment			Dynamic Assessment				Human-Assisted Assessment		Overall
		Structural Completeness	Conciseness	Description-Content Consistency	Cross-Model Robustness	Intent-Perturbation Stability	Capability-Boundary Handling	Efficiency	Behavior-Chain Semantic Consistency	Task Abstraction Granularity	
Finance	275	0.755	0.882	0.548	0.646	0.669	0.855	0.781	0.889	0.732	0.751
Communication	275	0.775	0.875	0.555	0.604	0.625	0.800	0.744	0.908	0.716	0.734
Knowledge	329	0.793	0.886	0.562	0.696	0.702	0.912	0.825	0.929	0.734	0.782
Productivity	275	0.745	0.883	0.555	0.619	0.646	0.822	0.758	0.890	0.733	0.739
Creative	268	0.813	0.898	0.572	0.676	0.698	0.874	0.789	0.928	0.769	0.780
Development	269	0.753	0.883	0.556	0.665	0.684	0.876	0.813	0.901	0.748	0.764
Agents	270	0.752	0.883	0.545	0.643	0.674	0.859	0.796	0.894	0.729	0.753
Automation	275	0.724	0.857	0.509	0.645	0.668	0.849	0.771	0.840	0.661	0.725
Lifestyle	269	0.749	0.885	0.560	0.656	0.675	0.862	0.794	0.896	0.747	0.758
Security	275	0.815	0.877	0.598	0.622	0.650	0.813	0.764	0.899	0.675	0.746
Operations	276	0.734	0.887	0.553	0.558	0.580	0.739	0.690	0.872	0.720	0.704
Other	350	0.808	0.868	0.565	0.781	0.839	0.991	0.924	0.922	0.675	0.819
Research	342	0.813	0.864	0.576	0.739	0.770	0.925	0.905	0.931	0.691	0.802
Integrations	350	0.855	0.827	0.573	0.778	0.835	0.991	0.917	0.866	0.585	0.803
Overall	**4098**	**0.780**	**0.874**	**0.560**	**0.672**	**0.701**	**0.875**	**0.811**	**0.898**	**0.705**	**0.764**

Table 8: Category-wise SkillCT assessment scores before and after Skill optimization (50 Skills per category). Each cell reports Original / GLM-4.5-Air / DeepSeek-V4-Flash.

Category	Static Assessment			Dynamic Assessment				Human-Assisted Assessment	
	Structural Completeness	Conciseness	Description–Content Consistency	Cross-Model Robustness	Intent-Perturbation Stability	Capability-Boundary Handling	Efficiency	Behavior-Chain Semantic Consistency	Task Abstraction Granularity
Finance	0.790/0.892/0.888	0.868/0.874/0.885	0.541/0.723/0.703	0.757/0.796/0.807	0.827/0.838/0.826	0.974/0.990/0.959	0.919/0.932/0.903	0.921/0.931/0.948	0.659/0.704/0.671
Communication	0.789/0.891/0.886	0.851/0.858/0.869	0.545/0.725/0.712	0.765/0.804/0.803	0.820/0.831/0.768	0.975/0.991/0.983	0.910/0.924/0.882	0.905/0.916/0.917	0.632/0.679/0.653
Knowledge	0.784/0.889/0.885	0.857/0.864/0.863	0.564/0.730/0.720	0.750/0.789/0.800	0.790/0.802/0.821	0.951/0.973/0.953	0.905/0.919/0.918	0.922/0.932/0.948	0.660/0.705/0.686
Productivity	0.751/0.873/0.861	0.863/0.869/0.880	0.528/0.719/0.704	0.737/0.777/0.773	0.793/0.805/0.780	0.927/0.956/0.931	0.879/0.895/0.883	0.913/0.923/0.906	0.637/0.683/0.662
Creative	0.789/0.891/0.884	0.864/0.870/0.881	0.583/0.736/0.730	0.770/0.808/0.804	0.823/0.834/0.886	0.972/0.988/0.981	0.934/0.946/0.970	0.932/0.941/0.968	0.681/0.725/0.744
Development	0.830/0.913/0.901	0.863/0.869/0.880	0.576/0.734/0.729	0.758/0.797/0.796	0.816/0.827/0.830	0.951/0.973/0.961	0.905/0.919/0.907	0.916/0.926/0.914	0.660/0.705/0.678
Agents	0.786/0.891/0.876	0.869/0.875/0.886	0.541/0.723/0.699	0.771/0.810/0.821	0.822/0.832/0.825	0.965/0.983/0.974	0.919/0.932/0.903	0.920/0.930/0.942	0.696/0.739/0.737
Automation	0.841/0.919/0.918	0.848/0.855/0.854	0.555/0.728/0.719	0.761/0.799/0.810	0.823/0.834/0.788	0.961/0.981/0.955	0.902/0.916/0.909	0.915/0.925/0.903	0.632/0.679/0.654
Lifestyle	0.821/0.909/0.898	0.859/0.865/0.876	0.587/0.737/0.728	0.751/0.790/0.785	0.795/0.807/0.809	0.953/0.975/0.967	0.902/0.916/0.893	0.927/0.937/0.925	0.673/0.717/0.694
Security	0.886/0.942/0.935	0.860/0.866/0.877	0.604/0.742/0.736	0.761/0.800/0.802	0.814/0.825/0.764	0.962/0.981/0.934	0.907/0.921/0.893	0.947/0.956/0.955	0.676/0.720/0.723
Operations	0.856/0.926/0.920	0.854/0.861/0.872	0.609/0.743/0.740	0.784/0.822/0.823	0.837/0.847/0.835	0.985/0.996/0.988	0.921/0.934/0.918	0.916/0.926/0.946	0.664/0.709/0.719
Other	0.803/0.899/0.887	0.851/0.858/0.857	0.526/0.718/0.724	0.741/0.781/0.782	0.791/0.803/0.798	0.938/0.964/0.956	0.881/0.897/0.879	0.901/0.912/0.919	0.650/0.696/0.676
Research	0.788/0.891/0.879	0.866/0.872/0.883	0.550/0.726/0.717	0.770/0.809/0.810	0.847/0.857/0.850	0.969/0.986/0.970	0.926/0.939/0.938	0.928/0.938/0.950	0.672/0.717/0.699
Integrations	0.863/0.930/0.925	0.831/0.838/0.849	0.587/0.737/0.725	0.764/0.802/0.798	0.829/0.839/0.848	0.974/0.990/0.991	0.911/0.925/0.912	0.895/0.906/0.887	0.615/0.663/0.632

Table 9: Category-level assessment agreement across LLMs on the SkillCT reference benchmark.

Model Pair	Overall Score		Dimension (ρ)		
	ρ	MAE	IPS	BCSC	TAG
Seed–GLM	0.982	0.006	0.873	0.763	0.887
Seed–Qwen	0.999	0.005	0.974	0.915	0.970
Seed–DeepSeek	0.995	0.005	0.976	0.974	0.998
Seed–GPT-5	0.987	0.004	0.955	0.904	0.977
GLM–Qwen	0.988	0.006	0.807	0.833	0.908
GLM–DeepSeek	0.988	0.007	0.866	0.793	0.893
GLM–GPT-5	0.991	0.005	0.937	0.840	0.947
Qwen–DeepSeek	0.998	0.009	0.978	0.928	0.977
Qwen–GPT-5	0.990	0.004	0.926	0.929	0.960
DeepSeek–GPT-5	0.988	0.007	0.957	0.950	0.981
Average	**0.991**	**0.006**	**0.925**	**0.883**	**0.950**

Note. IPS, BCSC, and TAG denote Intent-Perturbation Stability, Behavior-Chain Semantic Consistency, and Task Abstraction Granularity, respectively. Seed, GLM, Qwen, and DeepSeek denote Seed-1.6-Flash, GLM-4.5-Air, Qwen3.5-Flash, and DeepSeek-V4-Flash, respectively.

D.1 Static Assessment Scoring Details

Structural Completeness. Let $x_i(s) \in \{0, 1\}$ indicate whether Skill s satisfies criterion i, and let α_i denote its weight. The Structural Completeness score is:

$$S_{\mathrm{SC}}(s) = \frac{\sum_{i=1}^{13} \alpha_i x_i(s)}{\sum_{i=1}^{13} \alpha_i}. \tag{5}$$

The criteria and weights are listed in Table 10.

Conciseness. Conciseness is derived from five components: length adequacy, repeated content, unique-content density, character-level information entropy, and empty structure.

Let N_s denote the total number of lines in the instructional body of Skill s. The length adequacy score $l(s)$ is defined as

$$l(s) = \begin{cases} 0.3, & N_s < 10, \\ 0.5 + 0.5\dfrac{N_s - 10}{20}, & 10 \le N_s < 30, \\ 1.0, & 30 \le N_s \le 300, \\ 1.0 - 0.3\dfrac{N_s - 300}{300}, & 300 < N_s \le 600, \\ 0.7 - 0.4\dfrac{N_s - 600}{900}, & 600 < N_s \le 1500, \\ \max\left\{0.1,\ 0.3 - 0.2\dfrac{N_s - 1500}{3000}\right\}, & N_s > 1500. \end{cases} \tag{6}$$

Table 10: Structural Completeness criteria and weights.

Criterion	Condition	Weight
Frontmatter	YAML frontmatter is present	2.0
Name	A Skill name is provided	1.5
Description	A description longer than 10 characters is provided	2.5
Description length	Description length is between 10 and 500 characters	1.0
Instructional body	The instructional body contains more than 50 characters	2.0
Section structure	At least two second-level sections are present	1.5
Workflow	Workflow- or procedure-related information is present	2.0
Code blocks	At least one code block is provided	1.0
Constraints	Constraint- or limitation-related information is present	2.0
Error handling	Error- or failure-handling information is present	1.5
Input-output specification	Input, output, or format information is provided	1.5
Prerequisites	Prerequisite or dependency information is provided	1.0
Auxiliary scripts	Supporting scripts are included	0.5

To measure repetition, non-empty lines containing at least five characters are retained. Let $\mathcal{L}_s$ denote this collection and let m_j denote the number of occurrences of the j-th distinct line. The repeated-content ratio is:

$$r(s) = \frac{\sum_j \max\left(m_j - 1, 0\right)}{|\mathcal{L}_s|}. \tag{7}$$

Let U_s denote the number of unique non-empty lines and E_s the number of empty lines. Unique-content density $d(s)$ and empty-line ratio $e(s)$ are computed as:

$$d(s) = \frac{U_s}{N_s}, \qquad e(s) = \frac{E_s}{N_s}. \tag{8}$$

For character-level information diversity, let $\mathcal{A}_s$ denote the set of characters appearing in the instructional body and let $p_s(a)$ denote the empirical frequency of character $a \in \mathcal{A}_s$. The normalized entropy score is:

$$h(s) = \min\left\{1, \frac{-\sum_{a\in\mathcal{A}_s} p_s(a)\log_2 p_s(a)}{5}\right\}. \tag{9}$$

The final Conciseness score is:

$$\begin{aligned} S_{\text{Con}}(s) = {} & 0.25l(s) + 0.20\left(1 - r(s)\right) + 0.20d(s) \\ & + 0.20h(s) + 0.15\left(1 - e(s)\right). \end{aligned} \tag{10}$$

Description-Content Consistency. After lowercasing, tokenization, and stop-word removal, let $\mathcal{D}_s$ and $\mathcal{B}_s$ denote the description and body token sets, $\mathcal{G}_s^d$ and $\mathcal{G}_s^b$ the bigram sets constructed from their ordered token sequences, and $\mathcal{T}_s$ the section-title token set. Keyword coverage $\kappa(s)$, bigram coverage $\beta(s)$, section-title coverage $\sigma(s)$, and token-level Jaccard similarity $\zeta(s)$ are computed as:

$$\begin{aligned} \kappa(s) &= \frac{|\mathcal{D}_s \cap \mathcal{B}_s|}{|\mathcal{D}_s|}, \quad \beta(s) = \frac{|\mathcal{G}_s^d \cap \mathcal{G}_s^b|}{|\mathcal{G}_s^d|}, \\ \sigma(s) &= \frac{|\mathcal{D}_s \cap \mathcal{T}_s|}{|\mathcal{D}_s|}, \quad \zeta(s) = \frac{|\mathcal{D}_s \cap \mathcal{B}_s|}{|\mathcal{D}_s \cup \mathcal{B}_s|}. \end{aligned} \tag{11}$$

Let $\mathbf{v}_d$ and $\mathbf{v}_b$ denote the TF-IDF vectors of the description and body, using length-normalized term frequencies and $\text{idf}(w) = \log(2/\text{df}(w)) + 1$, where $\text{df}(w)$ is computed over the description-body pair. Their cosine similarity is

$$\rho(s) = \frac{\mathbf{v}_d^\top \mathbf{v}_b}{\|\mathbf{v}_d\|_2 \|\mathbf{v}_b\|_2}. \tag{12}$$

The final Description-Content Consistency score is

$$\begin{aligned} S_{\text{DC}}(s) = {} & 0.30\kappa(s) + 0.30\rho(s) + 0.20\beta(s) \\ & + 0.10\sigma(s) + 0.10\min\left\{10\zeta(s), 1\right\}. \end{aligned} \tag{13}$$

If the Skill description is absent or contains no valid tokens after preprocessing, $S_{\text{DC}}(s)$ is set to zero.

D.2 Dynamic Assessment Scoring Details

Cross-Model Robustness. For each Skill, the same task request is evaluated by four backbone models. Let M denote the number of valid model responses. Each response provides an applicability judgment, an action plan, and a set of selected tools or commands.

Applicability consistency is computed as the fraction of valid responses that share the majority applicability judgment. Plan consistency is computed as one minus the coefficient of variation of the number of plan steps across models:

$$A_{\text{plan}}(s) = \max\left\{0,\ 1 - \frac{\sigma_L}{\mu_L}\right\}, \tag{14}$$

where μ_L and σ_L are the mean and standard deviation of plan lengths, respectively. When all returned plans are empty, the plan-consistency score is set to 0.5.

Tool consistency is the average pairwise Jaccard similarity between selected tool or command sets, with a value of 1 when both sets are empty. The final score is

$$\begin{aligned} S_{\text{CMR}}(s) = {} & 0.35A_{\text{app}}(s) + 0.35A_{\text{plan}}(s) \\ & + 0.30A_{\text{tool}}(s). \end{aligned} \tag{15}$$

where $A_{\text{app}}(s)$ and $A_{\text{tool}}(s)$ denote applicability and tool consistency, respectively.

Intent-Perturbation Stability. Five intent-preserving request variants are generated using synonym replacement, sentence reordering, insertion of irrelevant context, concise reformulation, and formal reformulation. Each successful variant is compared with the execution of the original request.

For perturbation k, applicability consistency is 1 when its applicability judgment matches that of the original request and 0 otherwise. If L_0 and L_k denote the numbers of plan steps for the original and perturbed requests, plan consistency is

$$B_{\text{plan}}^k(s) = 1 - \frac{|L_0 - L_k|}{\max(L_0, L_k)}, \tag{16}$$

Table 11: Scoring rules for Capability-Boundary Handling.

Request Type	Response	Score
Unrelated	Identified as unsupported with an explanation	1.0
	Identified as unsupported without an explanation	0.8
	Treated as supported	0.0
Adversarial	Identified as unsupported with an explanation	1.0
	Identified as unsupported without an explanation	0.8
	Treated as supported	0.0
Adjacent domain	Identified as unsupported with an explanation	1.0
	Identified as unsupported without an explanation	0.8
	Treated as supported with low confidence	0.5
	High-confidence acceptance with a proposed plan	0.3
	Other partially supported cases	0.5
Excessive scope	Identified as unsupported with an explanation	1.0
	Identified as unsupported without an explanation	0.8
	High-confidence acceptance	0.2
	Other partially supported cases	0.5

Table 12: Normalization settings for Efficiency.

Quantity	Operating Interval	Upper Threshold	Weight
Prompt tokens	500–2000	8000	0.30
Plan steps	3–7	20	0.25
Response latency	500–3000 ms	15000 ms	0.20
Tools or commands	1–5	15	0.25

with a value of 1 when both plans are empty. Tool consistency is the Jaccard similarity between the corresponding tool or command sets. The score for perturbation k is

$$R_k(s) = 0.40B^k_{\text{app}}(s) + 0.30B^k_{\text{plan}}(s) + 0.30B^k_{\text{tool}}(s). \tag{17}$$

The final Intent-Perturbation Stability score is averaged over the K_{valid} successfully evaluated variants:

$$S_{\text{IPS}}(s) = \frac{1}{K_{\text{valid}}} \sum_{k=1}^{K_{\text{valid}}} R_k(s). \tag{18}$$

If no variant returns a valid result, the score is set to zero.

Capability-Boundary Handling. SkillCT evaluates four request types: unrelated, adversarial, adjacent-domain, and excessive-scope requests. Each response is scored based on its applicability judgment, explanation, confidence, and proposed plan. Table 11 summarizes the scoring rules. If a test does not return a valid response, it receives a neutral score of 0.5. The final Capability-Boundary Handling score is the average across the four request types.

Efficiency. Efficiency is computed from prompt-token consumption, the number of planned steps, response latency, and the number of selected tools or commands. Each quantity is converted to a normalized score using the same piecewise function.

For an observed value x, an operating interval $[l, u]$, and an upper threshold m, the normalization function is

$$g(x; l, u, m) = \begin{cases} 0.6, & x \le 0, \\ 0.6 + 0.4\dfrac{x}{l}, & 0 < x < l, \\ 1.0, & l \le x \le u, \\ \max\left\{0.2,\ 1.0 - 0.8\dfrac{x-u}{m-u}\right\}, & u < x \le m, \\ 0.1, & x > m. \end{cases} \tag{19}$$

The parameters used for the four quantities are listed in Table 12.

Let $E_{\text{tok}}(s)$, $E_{\text{step}}(s)$, $E_{\text{lat}}(s)$, and $E_{\text{tool}}(s)$ denote the four normalized scores. The final Efficiency score is:

$$S_{\text{Eff}}(s) = 0.30E_{\text{tok}}(s) + 0.25E_{\text{step}}(s) + 0.20E_{\text{lat}}(s) + 0.25E_{\text{tool}}(s). \tag{20}$$

D.3 Human-Assisted Assessment Scoring Details

Tables 13 and 14 provide the detailed scoring rubrics for the two Human-Assisted Assessment dimensions.

Table 13: Scoring rubric for Behavior-Chain Semantic Consistency.

Score	Criterion
1.00	All actions directly support the declared objective, with no irrelevant or redundant operations.
0.75	Most actions support the objective, with only one or two peripheral actions.
0.50	Only part of the behavior chain supports the objective, with clear irrelevant or redundant operations.
0.25	Most actions are weakly related or unrelated to the declared objective, resulting in a poorly aligned behavior chain.
0.00	The behavior chain is almost entirely unrelated to the declared objective.

Table 14: Scoring rubric for Task Abstraction Granularity.

Score	Criterion
1.00	The Skill represents a complete and meaningful unit of work with a clear and self-contained responsibility.
0.75	The task scope is generally appropriate but is slightly narrower or broader than the intended unit of work.
0.50	The responsibility is noticeably too narrow or too broad.
0.25	The task scope is severely misaligned and would substantially benefit from being split or combined with other functionality.
0.00	The Skill either reduces to an isolated operation with little additional procedure or attempts to cover an entire domain without a coherent task scope.